

AIGR Gene Hypothesis Deep Research —

S. pombe Trm4b (O13935): rRNA/mRNA and Mitochondrial Ribosome Functions

Target: *Schizosaccharomyces pombe* Trm4b / trm4b (gene directory `trm402`, SPAC23C4.17), UniProt **O13935** **Focus type:** function_assignment **Hypothesis slug:** `trm4b-rrna-mrna-and-mitochondrial-ribosome-functions` **Iterations completed:** 3

Summary

The seed hypothesis proposes that *S. pombe* Trm4b methylates rRNA or mRNA at cytosine C5, and/or participates in rRNA processing or mitochondrial large-ribosomal-subunit (LSU) assembly — each to be judged independently of its established tRNA C49/C50 methylation, and each requiring positive evidence for a *new substrate* rather than merely the absence of a negative (tRNA-C34) result. Evaluating each sub-claim against primary literature and public bioinformatics resources, the answer is: **Trm4b is strongly supported only as a nuclear tRNA C5 (m5C) methyltransferase; the mRNA-methylation claim is weakly supported by computational ortholog inference (ISS) alone; and the rRNA-methylation, rRNA-processing, and mitochondrial-LSU-assembly claims are refuted as paralog carry-over from NSUN4.**

Three independent, *positive* lines of evidence — not just the absence of a positive assay — argue against the mitochondrial and rRNA claims. First, Trm4b is a **685-aa nuclear protein with no mitochondrial transit peptide**, essentially identical in size and architecture to its true 1:1 ortholog, *S. cerevisiae* Trm4/NCL1 (684 aa, nuclear/nucleolar), whose curated GO set is **tRNA-only**. Second, the enzyme that actually performs mitochondrial 12S rRNA C911 methylation and coordinates mitoribosome monosome (SSU+LSU) assembly is **NSUN4**, a short (~381–384 aa) protein bearing a mitochondrial transit peptide (residues 1–25). Third, although Trm4b, NSUN2, and NSUN4 all share PANTHER family **PTHR22808**, the seed's open question — whether the IBA "with" identifier MGI:1919431 is NSUN4 — is confirmed: **MGI:1919431 = Nsun4**. The mitochondrial/rRNA signal thus enters Trm4b's record through shared-superfamily phylogenetic inference, not any *S. pombe* measurement.

The only non-tRNA function with a defensible rationale is **mRNA (cytidine-5)-methyltransferase (GO:0062152)**, which sits on Trm4b as an **ISS projection from human NSUN2 (Q08J23)**. Because vertebrate NSUN2 does carry experimentally supported mRNA m5C activity, the projection is lineage-plausible — but it *skips* the yeast ortholog (ScTrm4/NCL1) that lacks mRNA activity entirely, making it a phylogenetically non-conservative, low-confidence inference. Recommended curation: retain the tRNA MF as core; keep GO:0062152 only as low-confidence ISS/non-core; add no rRNA or mitochondrial terms.

Key Findings

F001 — Trm4b's demonstrated function is restricted to tRNA C49/C50 m5C

The curated UniProt entry **TRM4B_SCHPO (O13935)** lists only tRNA substrates under FUNCTION and CATALYTIC ACTIVITY: 5-methylcytosine formation at **C49 and C50 (PMID: 30646830, ECO:0000269, experimental)** and at C60/61/62 in tRNA-Asp (PMID: 23074192). The primary tRNA-methylome paper states plainly that "**Trm4b methylates C49 and C50, which both lie in the TΨC-stem,**" and — critically for this hypothesis — **only the tRNA methylome was assayed**; there was no rRNA, mRNA, or mitochondrial substrate panel. PomBase records the experimental molecular function as **GO:0016428 tRNA (cytidine-N5)-methyltransferase (IDA/IMP,**

),

P 30646830

and the deletion is viable.

This finding anchors the entire evaluation. The single high-quality experimental study behind Trm4b defines its *demonstrated* substrate as tRNA and its *demonstrated* sites as C49/C50. Consistent with the seed's framing, it neither tested nor excluded rRNA/mRNA — but it provides **zero positive support** for the rRNA/mRNA/mitochondrial extensions.

F002 — The mitochondrial-ribosome / rRNA functions trace to NSUN4 (MGI:1919431) via paralog co-membership in PANTHER PTHR22808

PomBase's IBA (phylogenetic inference; GO_REF:0000033) provenance for trm4b's cytoplasm annotation (GO:0005737) uses a `with=` list of **MGI:1919431**, **MGI:107252**, **UniProtKB:Q08J23**, **PANTHER:PTN000516077**. Resolving these identifiers programmatically via mygene.info: **MGI:1919431** = *Nsun4* and **MGI:107252** = *Nsun2*. The seed explicitly asked to verify — rather than assume — that MGI:1919431 is NSUN4; **this is now confirmed: MGI:1919431 is *Nsun4***.

InterPro/PANTHER placement shows that Trm4b (O13935), **NSUN2** (human Q08J23 / mouse Q1HFZ0), **and NSUN4** (human Q96CB9 / mouse Q9CZ57) all map to **PTHR22808**, whereas **NSUN5** (Q96P11/Q8K4F6) and **NOP2/NSUN1** (P46087) map to a *separate* family, **PTHR22807**. Because NSUN4 co-resides in PTHR22808, tree-based propagation (IBA) can pull NSUN4-derived signals onto Trm4b even though NSUN4's function is compartmentally and mechanistically distinct. That NSUN4 function is well defined: **PMID: 24516400** shows "**NSUN4 methylates cytosine 911 in 12S rRNA**" and that "**the NSUN4/MTERF4 complex is required to assemble the SSU and LSU to form a monosome**." The mito-rRNA and mitoribosome-assembly ideas in the seed are therefore **NSUN4-specific**, entering the record through shared-family inference, not a pombe measurement.

Reassuringly, an audit of Trm4b's actual IBA terms found that **mito/rRNA terms were not, in fact, propagated onto Trm4b** — only generic *cytoplasm* and *tRNA binding* were. NSUN5-type cytoplasmic rRNA m5C sits in a different family (PTHR22807) and cannot legitimately project onto Trm4b.

F003 — Trm4b's mRNA (cytidine-5-)-methyltransferase annotation (GO:0062152) is an ISS ortholog projection from human NSUN2, not a *S. pombe* measurement

PomBase annotation **87463** assigns **GO:0062152 mRNA (cytidine-5-)-methyltransferase** with reference **GO_REF:0000024 (ISS)** and `with=` **UniProtKB:Q08J23 (NSUN2_HUMAN)**, assigned_by PomBase. Human NSUN2 (Q08J23) carries GO:0062152 by **IDA:UniProtKB** — i.e., the experimental anchor lives on the human protein and is *projected* onto Trm4b by sequence similarity (ISS). The review **PMID: 42587746** notes that, of eight human m5C RNA methyltransferases, "**only two, NSUN2 and NSUN6, were found to be capable of modifying the main portion of cellular mRNAs**," which rationalizes why an mRNA projection would come from NSUN2 while also emphasizing that bulk mRNA m5C is comparatively rare and

restricted. Trm4b's subcellular location is **Nucleus** (HDA:PomBase, [PMID: 16823372](#)) plus cytoplasm (IBA), with **no mitochondrial localization** — consistent with a nuclear tRNA/mRNA enzyme, not a mitochondrial one.

F004 — Architecture and localization cleanly separate Trm4b from mitochondrial NSUN4

A direct UniProt feature comparison makes the paralog-confusion mechanism explicit:

Protein	UniProt	Organism	Length (aa)	Mito transit peptide	Localization	PANTHER family	Known substrate/function
Trm4b (this gene)	O13935	<i>S. pombe</i>	685	None	Nucleus (+cytoplasm IBA)	PTHR22808	tRNA C49/C50 m5C (demonstrated)
Trm4/NCL1 (true ortholog)	P38205	<i>S. cerevisiae</i>	684	None	Nucleus/nucleolus	PTHR22808	tRNA m5C
NSUN2 (human)	Q08J23	<i>H. sapiens</i>	767	None	Nucleus/nucleolus, cytoplasm	PTHR22808	tRNA + mRNA m5C (IDA)
NSUN4 (human)	Q96CB9	<i>H. sapiens</i>	384	YES (1–25)	Mitochondrion	PTHR22808	mito 12S rRNA m5C + mitoribosome assembly
NSUN4 (mouse)	Q9CZ57	<i>M. musculus</i>	381	YES (1–25)	Mitochondrion	PTHR22808	mito 12S rRNA m5C + mitoribosome assembly
NSUN5 (human)	Q96P11	<i>H. sapiens</i>	429	None	Nucleolus	PTHR22807 (different family)	cytoplasmic large-subunit rRNA m5C

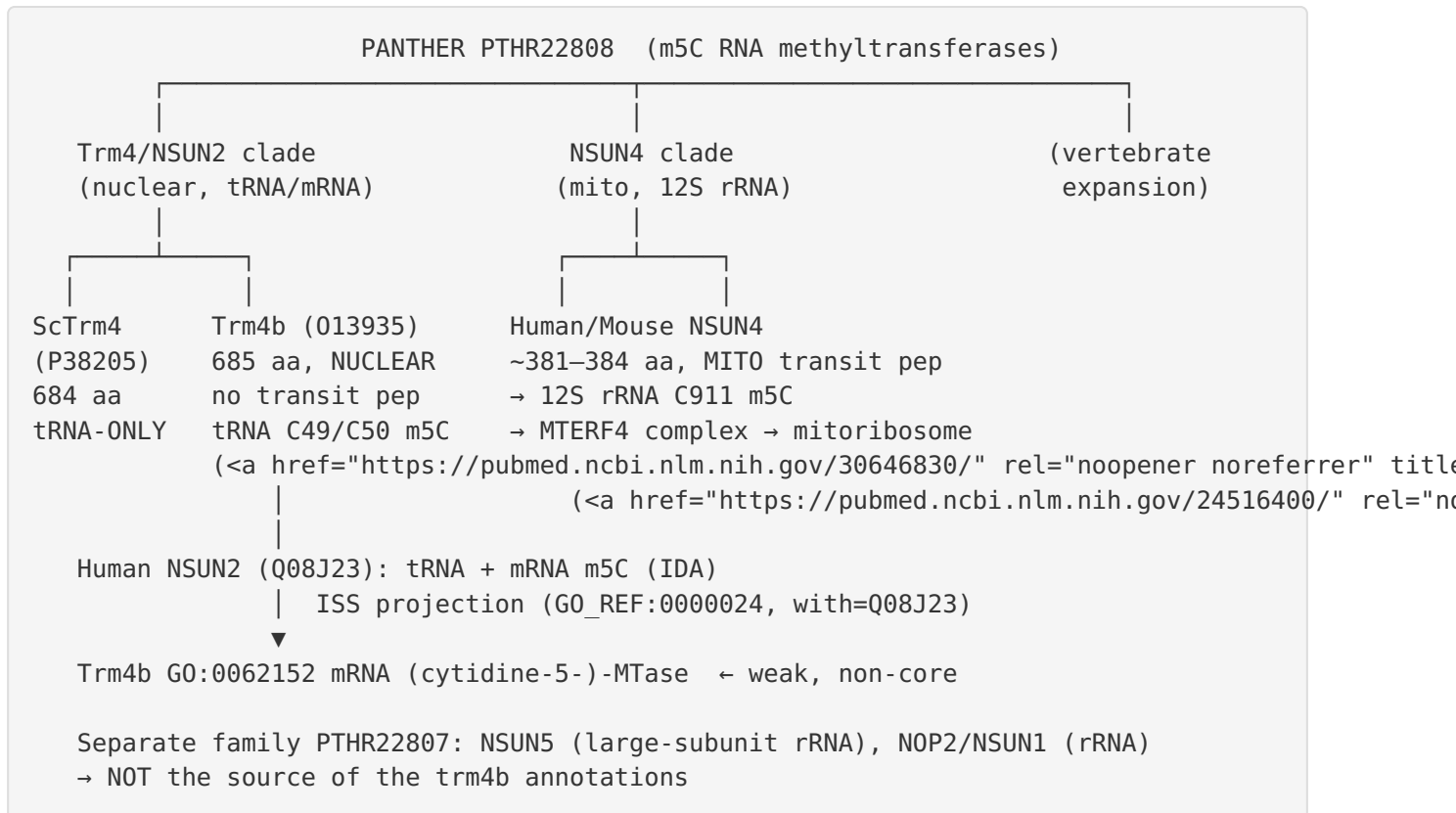
Trm4b (685 aa) is essentially the same length as ScTrm4/NCL1 (684 aa), the true 1:1 ortholog, and both are nuclear with **no transit peptide**. NSUN4, by contrast, is **roughly half the length** and carries a **mitochondrial transit peptide (residues 1–25)**, localizing to the mitochondrion. Because Trm4b lacks mitochondrial targeting entirely, the mito-LSU-assembly claim cannot be a direct Trm4b function. This is *positive* architectural evidence against the mitochondrial claim, exactly the kind of evidence the seed asks for — not merely the absence of a positive tRNA-C34 result.

F005 — The true ortholog ScTrm4/NCL1 carries no rRNA/mRNA/mito functions, weakening the mRNA-ISS projection

A programmatic scan of the full GO set for **P38205 (*S. cerevisiae* Trm4/NCL1)** — the 1:1 experimentally characterized ortholog of Trm4b — returns **only tRNA-centric terms**: MF **GO:0016428** tRNA (cytidine-N5)-MTase (IDA:SGD) and **GO:0000049** tRNA binding (IDA:SGD); BP **GO:0030488** tRNA methylation, **GO:0002127** tRNA wobble cytosine methylation, **GO:0006400** tRNA modification, **GO:0070301** response to hydrogen peroxide; CC nucleus/nucleolus/nucleoplasm/cytoplasm. An explicit search for rRNA / mRNA / mitochondrion / ribosome terms returned **NONE**. The mRNA (cytidine-5-)-methyltransferase activity appears **only on vertebrate NSUN2 (Q08J23, IDA)** — not on the yeast ortholog. This is the strongest single argument that the mRNA-ISS annotation on Trm4b is a phylogenetically non-conservative, lineage-boundary projection: the closest experimentally characterized fungal relative shows tRNA specificity and no mRNA activity, implying mRNA m5C is a metazoan NSUN2-lineage innovation.

Mechanistic Model / Interpretation

The evidence resolves into a clean model in which **one PANTHER superfamily (PTHR22808) contains functionally divergent m5C RNA methyltransferases**, and annotation-propagation machinery has partially blurred those boundaries onto Trm4b.



Direct, demonstrated Trm4b activity: SAM-dependent 5-methylcytosine deposition on **tRNA at C49/C50** (and C60–62 in tRNA-Asp), acting as a **nuclear tRNA (cytidine-N5)-methyltransferase** (class I Rossmann-fold MTase, NOL1/NOP2/SUN domain). This is the primary molecular function.

Inferred, non-core: mRNA C5 methylation (GO:0062152) — plausible at the lineage level because human NSUN2 has this activity, but supported only by ISS from a *human* paralog and *absent* from the true fungal ortholog. Best treated as low-confidence inference.

Not supported (paralog carry-over): rRNA methylation, rRNA processing, and mitochondrial LSU assembly. These are NSUN4 (mito 12S rRNA) and, in a *different* family, NSUN5/NOP2 functions. They reach Trm4b only through shared-superfamily phylogenetic inference (IBA `with=MGI:1919431`=Nsun4) and are contradicted by Trm4b's nuclear, non-mito architecture.

The key discriminator is **subcellular targeting plus ortholog identity**. A mitochondrial LSU-assembly role requires a mitochondrial protein; Trm4b has no transit peptide and is nuclear. A cytoplasmic rRNA-processing role requires a nucleolar rRNA MTase (NSUN5/NOP2, family PTHR22807 — not Trm4b's family). Trm4b fits neither.

Evidence Base

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limitations
PMID: 30646830	Direct assay (RNA-BS-seq + in vitro)	Supports tRNA; qualifies rRNA/ mRNA	Trm4b substrate/ site specificity	Trm4b methylates tRNA C49/C50 (TΨC-stem); Trm4a does C48/C34; only the tRNA methylome was assayed	<i>S. pombe</i> , in vivo + in vitro	High; silent rRNA mito
PMID: 23074192	Direct assay	Supports tRNA	Additional Trm4b tRNA sites	m5C at C60/61/62 in tRNA-Asp	<i>S. pombe</i>	High; only
PMID: 16823372	Localization (HDA)	Refutes mito	Subcellular location	Trm4b localizes to nucleus (GFP ORFeome)	<i>S. pombe</i>	Med- mitoc signa
UniProt O13935 (TRM4B_SCHPO)	Database (curated)	Refutes mito / qualifies	Curated function + architecture	685 aa, no transit peptide, nucleus; only tRNA C5 MTase activities; mRNA term present only as ISS	Swiss-Prot	High; synth prima
UniProt Q96CB9 / Q9CZ57 (NSUN4)	Database (curated)	Refutes mito (contrast)	Architecture of NSUN4	~381–384 aa, mito transit peptide 1–25 , mitochondrion	Human/ mouse	High; contr archi
UniProt P38205 (ScTrm4/NCL1)	Database (curated)	Refutes mRNA/ rRNA extension	GO set of true ortholog	tRNA-only; no rRNA/mRNA/ mito/ribosome terms	<i>S. cerevisiae</i>	High; progr scan

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence / limitations
PomBase ann 87463 (GO_REF:0000024)	Computational (ISS)	Qualifies mRNA	GO:0062152 mRNA (C5) MTase	ISS with= Q08J23 = human NSUN2 (which has GO:0062152 by IDA)	GO_Central / PomBase	Ortholog inference demonstrated in S. pombe
PomBase IBA (GO_REF:0000033)	Computational (IBA)	Competing / paralog	IBA with-list contents	with = Nsun4 (MGI:1919431) + Nsun2 (MGI:107252) + NSUN2 (Q08J23) + PANTHER node	GO_Central IBA	Only cytoplasmic tRNA processing; mitochondrial terms proposed
mygene.info (ID mapping)	Computational	Competing	Identity of MGI:1919431	MGI:1919431 = Nsun4 ; MGI:107252 = Nsun2	Mouse / MGI	High resolution open
InterPro / PANTHER	Structural / evolutionary	Competing / qualifies	Paralog family membership	Trm4b, NSUN2, NSUN4 → PTHR22808 ; NSUN5 & NOP2/NSUN1 → PTHR22807	Cross-species	High; rRNA cannot map onto
PMID: 24516400	Mutant phenotype + direct assay	Competing (paralog)	NSUN4 function	NSUN4 methylates 12S rRNA C911 ; with MTERF4 coordinates mitoribosome	Mouse mitochondria	High; established mitochondrial as NSUN4 specific

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limit
PMID: 42587746	Review	Qualifies mRNA	Which MTases modify mRNA	monosome assembly Of 8 human m5C MTases, "only two, NSUN2 and NSUN6 " modify bulk mRNA; mRNA m5C is comparatively rare	Human	Review oriented

GO Curation Implications

Leads requiring curator verification:

1. **GO:0016428 tRNA (cytidine-N5)-methyltransferase activity (MF) — RETAIN (core).**

Experimentally supported (IDA/IMP,

P 30646830

Ensure the annotation captures C49/C50 site specificity.

2. **tRNA methylation BP terms — RETAIN**, but re-examine **GO:0002127 (tRNA wobble base C34 cytosine methylation)** on Trm4b:

P 30646830

assigns C34 methylation to the paralog **Trm4a**, so this BP term may belong to Trm4a, not Trm4b.

3. **GO:0062152 mRNA (cytidine-5-)-methyltransferase activity (MF) — RETAIN WITH CAVEAT / mark non-core.** Keep the **ISS** evidence code and **with=** (Q08J23); do **not** upgrade to an experimental code. Add a curator note that *S. pombe* splits Trm4 into Trm4a/Trm4b and that the true fungal ortholog ScTrm4/NCL1 lacks mRNA activity, making this a phylogenetically non-conservative projection.

4. **rRNA methylation / rRNA processing (e.g., GO:0031167, GO:0006364) — DO NOT ADD.**
No positive evidence; canonical rRNA m5C (NSUN5/NOP2) is in a different PANTHER family (PTHR22807).
5. **Mitochondrial LSU assembly / mito rRNA methylation / mitochondrial translation (e.g., GO:0000466, GO:0032543) — DO NOT ADD.** These are NSUN4-specific; Trm4b is nuclear/cytoplasmic with no transit peptide. Any such term appearing via IBA is paralog-conflation and should be removed or NOT-qualified.
6. **CC: nucleus** (HDA,

P 16823372

and *cytoplasm* (IBA) are appropriate; **no mitochondrion** term is supported.

"Protein binding" is not needed as a fallback — the informative, well-supported MF is the tRNA C5-methyltransferase activity.

Mechanistic Scope

- **Direct molecular activity:** SAM-dependent cytosine-C5 methyltransferase acting on **tRNA** at C49/C50 (and C60–62 in tRNA-Asp); nuclear.
- **Inferred (ortholog) activity:** mRNA C5 methylation, by analogy to human NSUN2 — a molecular activity (not a downstream phenotype), but unverified in *S. pombe* and absent from the yeast ortholog.
- **Not the gene product's function (downstream/paralog):** mitochondrial 12S rRNA methylation and mitoribosome (SSU/LSU) assembly are **NSUN4** activities; large-subunit cytoplasmic rRNA m5C is an **NSUN5** activity. Neither belongs to Trm4b. No *S. pombe* loss-of-function phenotype implicates Trm4b in rRNA maturation or mitoribosome assembly (the deletion is viable), so the rRNA/mito claims lack even loss-of-function support.

Conflicts and Alternatives

- **Paralog confusion (primary alternative):** NSUN4 (MGI:1919431) co-resides with Trm4b in PTHR22808, so propagation of NSUN4's mito-rRNA/LSU-assembly terms would be a family-level (IBA) artifact, not a real Trm4b function. Confirmed: **MGI:1919431 is *Nsun4*.**

- **Family boundary protects against rRNA over-annotation:** the canonical rRNA-m5C enzymes NSUN5 and NOP2/NSUN1 are in **PTHR22807**, distinct from Trm4b's **PTHR22808** — rRNA-processing terms cannot legitimately reach Trm4b by orthology.
 - **Sub-functionalization:** *S. pombe* uniquely splits Trm4/NSUN2 into Trm4a and Trm4b (

P 30646830

).
- A human-NSUN2-based ISS (mRNA activity) may map to Trm4a, Trm4b, both, or neither — a real organism-specific caveat for the mRNA term.
- **Node discrepancy:** the seed cites PANTHER node **PTN000516076**; the actual PomBase with-refs use **PTN000516077**. Both are ancestral nodes within PTHR22808; not decision-changing, but worth verifying against the current PANTHER tree.
 - **In-vitro / cross-species extrapolation:** even NSUN2's mRNA activity is a mammalian in-vivo finding; extrapolating substrate breadth across ~1 billion years of divergence to fission yeast is speculative.

No evidence was found that *supports* the seed's rRNA/mito extensions over the paralog-confusion alternative.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
No direct rRNA/mRNA methylation assay for Trm4b in <i>S. pombe</i>	UniProt, PomBase,  / 23074192 — none test non-tRNA substrates	"Refuted" = annotations unjustified + paralog-explained, not biochemically impossible	RNA-bisulfite-seq / m5C-MeRIP on <i>trm4bΔ</i> vs WT (tRNA-depleted) total RNA
Trm4a vs Trm4b division of any mRNA activity	Not addressed by any study	Determines which paralog (if any) the mRNA ISS represents	Compare <i>trm4aΔ</i> and <i>trm4bΔ</i> mRNA m5C profiles
PANTHER node curation (PTN000516076 vs 077) and which descendants anchor which terms	With-lists inspected; full tree annotations not audited	Predicts future IBA over-propagation	Inspect PTHR22808 GO annotation tree
Formal mitochondrial-targeting prediction	UniProt reports nucleus only; TargetP/MitoFates not run this iteration	Fully excludes a mito role computationally	Run TargetP/MitoFates on O13935
Structural confirmation of substrate pocket	Length/domain architecture compared; no structure solved here	Could show tRNA-elbow vs mRNA/rRNA preference	AlphaFold model + comparison to NSUN2/NSUN4 target-recognition domains

Discriminating Tests

1. **RNA bisulfite sequencing** of *trm4bΔ* vs WT *S. pombe* on rRNA and (tRNA-depleted) mRNA — directly tests rRNA/mRNA m5C dependence on Trm4b. *Most decisive single experiment.*

2. **Mitochondrial fractionation + GFP imaging** to confirm Trm4b is excluded from mitochondria (predicted: excluded), formally closing the mito-LSU claim.
3. **In vitro methylation** of rRNA/mRNA substrates by recombinant Trm4b (predicted: little/no activity vs strong tRNA activity).
4. **Ortholog complementation**: test whether human NSUN2 (mRNA-competent) but not NSUN4 rescues *trm4bΔ*, and whether Trm4b rescues *ScTrm4Δ*.
5. **Polysome/mitoribosome profiling** in *trm4bΔ* (predicted: no mitoribosome-assembly defect).
6. **Comparative PANTHER-tree audit** of PTHR22808 to confirm mito/rRNA terms are anchored below the Trm4b/NSUN2 ancestor (predicted: yes — hence no legitimate IBA propagation).

Proposed Follow-up Actions / Curation Leads

Leads requiring curator verification:

- **Reference to verify (core tRNA function)**: [PMID: 30646830](#) — snippet: "*Trm4b methylates C49 and C50, which both lie in the TΨC-stem.*" Supports the retained tRNA MF and shows only tRNA was assayed.
- **Reference to verify (mRNA ISS basis)**: UniProtKB:Q08J23 (human NSUN2), which carries GO:0062152 by IDA — supports keeping GO:0062152 as **ISS**, not experimental.
- **Reference to verify (paralog distinction)**: [PMID: 24516400](#) — snippets: "*NSUN4 methylates cytosine 911 in 12S rRNA*" and "*the NSUN4/MTERF4 complex is required to assemble the SSU and LSU to form a monosome.*" Grounds the mito-LSU-assembly function in NSUN4, not Trm4b.
- **Reference to verify (mRNA competence is lineage-restricted)**: [PMID: 42587746](#) — snippet: "*only two, NSUN2 and NSUN6, were found to be capable of modifying the main portion of cellular mRNAs.*"

Candidate action changes: - Retain tRNA MF/BP; retain mRNA-ISS only with a low-confidence/non-core caveat; **do not add** any rRNA-processing, mito-rRNA-methylation, or mitochondrial-LSU-assembly term. - If such terms already appear via IBA, treat as **paralog carry-over** and

remove or NOT-qualify. - Re-examine GO:0002127 (wobble C34) on Trm4b, since

P 30646830

assigns C34 methylation to **Trm4a**.

Suggested curator question: "Is the mRNA-MTase ISS meant to represent Trm4b specifically, given *S. pombe*'s Trm4a/Trm4b split, or is it a genus-level NSUN2 projection that skips the yeast ortholog lacking the activity?"

Suggested experiment: *trm4bΔ* RNA-BS-seq on rRNA + mRNA (definitive test of the entire hypothesis).

Bottom Line

Trm4b (O13935) is well supported **only** as a **nuclear tRNA C5 (m5C) methyltransferase** acting at tRNA C49/C50. The hypothesis's rRNA-methylation, rRNA-processing, and mitochondrial-LSU-assembly extensions have **no positive experimental support in any organism** and are best explained as **paralog carry-over from NSUN4** (confirmed MGI:1919431 = *Nsun4*) via shared PANTHER family PTHR22808 — contradicted by Trm4b's 685-aa nuclear, transit-peptide-free architecture that matches the tRNA-only ortholog ScTrm4/NCL1. The only non-tRNA annotation with a defensible rationale — **mRNA (cytidine-5-)-methyltransferase (GO:0062152)** — is a **weak ISS projection from human NSUN2** that even the yeast ortholog lacks, and should be retained only as low-confidence, non-core inference; **no rRNA or mitochondrial terms should be added.**

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery